

claude-windows / c1fcb4aa-537d-41fd-858e-53f93349bf5a

Metadata

```
- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\subagents\workflows\wf_7049e1ed-b93\agent-a8fe07ff859f4e642.jsonl`
- SHA-256: `31d02363387d35e7f93fe1c73168a36d9a06739c8ac63252e72100408ece0e6a`
- Source modified: `2026-07-02T04:51:10+00:00`
- Imported at: `2026-07-05T16:48:24+00:00`
- project: `wf_7049e1ed-b93`
- session_id: `c1fcb4aa-537d-41fd-858e-53f93349bf5a`
```

Transcript

1. user (2026-07-02T04:45:39.423Z)

You are a senior engineer producing ONE isolated PR. Today is 2026-07-02.

REPO: C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer (a shared checkout ? follow the safety rules exactly).

AUDIT FINDINGS YOU ARE FIXING: R2-5 (CLAUDE.md 3 weeks false) ? Grep

C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/AUDIT_REPORT_khelsutra-guru_2026-07-02.md for these IDs and read the full evidence + verifier notes before coding.

TASK BRIEF:

In C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer: rewrite the stale 'Current state' section of CLAUDE.md (lines ~30-42, last true 2026-06-10). It falsely claims no platform/owned-model work started, calls the async job model 'committed next slice', and marks StorageBackend/ComputeBackend 'Future' ? all shipped (backend/api/job_worker.py, backend/storage/, backend/compute/, flywheel M11, gdrive-api M12, compute-picker, async compile v9 schema...). VERIFY every sentence you write against code/git log (cite PRs in the text where the file already does). PRESERVE all owner-gate language that is still true (owned-model TRAINING/greenlight gating may still be owner-gated ? check docs/NEXT_STEPS.md's NO-GO block and keep consistent). Keep the section's format/length discipline; this file is agent-loaded memory ? accuracy over completeness. Docs-only PR.

BRANCHING SAFETY (mandatory ? shared checkout, sibling sessions exist):

1. In the MAIN repo checkout run: git fetch origin
2. Create an ISOLATED WORKTREE: git worktree add "C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-<shortname>" -b <branch-name> origin/master (branch explicitly FROM origin/master ? never from the current local branch; use origin/main for repos whose default is main)
3. Do ALL work inside the worktree. Before committing: git branch --show-current + git log --oneline -1 must show YOUR branch on the origin-master base.
4. Stage EXPLICIT paths only (git add <file> <file>) ? NEVER git add -A / . / -u
5. Before opening the PR: git log --oneline origin/master..HEAD must list ONLY your commits.

COMMIT/PR CONVENTIONS:

- Commit message: conventional style matching the repo's history; end the message with a blank line then: Co-Authored-By: Claude Fable 5 <noreply@anthropic.com>
- PR body: what/why, a "## Verification" section stating exactly what you ran locally and what CI covers, and end with: ? Generated with [Claude Code](https://claude.com/claude-code)
- Do NOT merge the PR. Do NOT remove the worktree (a fixer may need it). Return the PR number, branch, worktree path.

QUALITY BAR:

- Read the repo's CLAUDE.md first; for the engine also docs/CODE_MAP.md §0 (file placement) before adding any new file.
- Add/extend tests for every behavior change. Run the targeted tests you touched (python -m pytest tests/<relevant> -q) + ruff check on changed files; if mypy config exists, run mypy on touched modules. Full suite is CI's job ? do not run it.
- Default-safe: new knobs default OFF / behavior-preserving unless the task explicitly says

otherwise.

- The full audit evidence is in C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/AUDIT_REPORT_khelsutra-guru_2026-07-02.md ? Grep it for your finding IDs and READ the verifier notes; they contain corrections (exact file:line, scoping traps) you MUST honor.

If the brief's premise turns out false against current code (something already fixed, file moved), adapt honestly and say so in notes ? never force a pointless diff. If you genuinely cannot complete (env broken, auth missing), open NO PR and explain in notes with pr_number=0.

2. user (2026-07-02T04:45:46.995Z)

450-

451-[Omitted long context line]

452-

453-(4) No fix landed: git log for content.ts/panel.ts/directVideo.ts shows only ad6ac9c (initial), 5ec1831 and 0a05ecf (i18n) ? no gating change. No visibilitychange/IntersectionObserver/MutationObserver/activeTab anywhere in web/ (grep: only hit is the setInterval itself).

454-

455-(5) Intent: ``<all_urls>`` is deliberate (DESIGN.md LD-7: panel hidden by default, icon is the explicit "annotate this page" gesture) ? but LD-7 gates only the UI, not the scan loop, so the perf cost contradicts the design's own "not intrusive" rationale.

456-[Omitted long context line]

457-

458-Severity P2 not P1: no correctness/data-loss/security impact, and the extension is version 0.1.0/private (not store-published), so blast radius today is dev machines; would be P1 if shipped broadly.

459-

460-[Omitted long context line]

461-

462-### R2-2 [P2/S] ? rally-annotator

463-****Locked decision LD-i8 (web language persisted + seeded from browser language) is documented as shipped but not implemented ? panel always boots in English and forgets the choice on reload****

464-- verdict: CONFIRMED

465-[Omitted long context line]

466-[Omitted long context line]

467-

468-### R2-3 [P2/S] ? sports-data-collector

469-****Collector pins obsreport as a plain PyPI requirement while the name is unclaimed on PyPI ? dependency-confusion window and install broken today****

470-- verdict: CONFIRMED

471-[Omitted long context line]

472-[Omitted long context line]

473-

474-### R2-4 [P2/S] ? badminton-highlight-indexer

475-****No timeout on any serving-path ffmpeg subprocess ? one hung ffmpeg permanently wedges the single-worker job queue****

476-- verdict: CONFIRMED

477-[Omitted long context line]

478-[Omitted long context line]

479-

480-### R2-5 [P1/S] ? badminton-highlight-indexer

481-****Engine CLAUDE.md 'Current state' section is ~3 weeks false ? claims platform/owned-model work 'has not started'****

482-- verdict: CONFIRMED

483-[Omitted long context line]

484-[Omitted long context line]

485-

486-### R2-6 [P1/M] ? sports-data-collector

487-****Golden-corpus registry rot: all 6 owned 'curated' videos point at a deleted absolute local_path and the 4.9GB source videos are single-copy with no documented backup****

488-- verdict: CONFIRMED

489-[Omitted long context line]
490-[Omitted long context line]
491-
492-### R2-7 [P2/S] ? non-git
493-**Roora vendor pilot record (paid deliverables + corrected golden labels + 3 videos) is a single unversioned local copy at risk of loss**
494-- verdict: CONFIRMED
495-[Omitted long context line]
496-[Omitted long context line]
497-
498-### R2-8 [P1/S] ? badminton-highlight-indexer
499-**Upload retention sweep tool exists but nothing schedules it ? unbounded upload growth + unenforced 7-day privacy retention on any hosted deployment (#301)**
500-- verdict: CONFIRMED
501-[Omitted long context line]
502-[Omitted long context line]
503-
504-
505-## Owner-gated (flagged, no action)
506-
507-[Omitted long context line]
508-
509-[Omitted long context line]
510-

3. user (2026-07-02T04:45:47.799Z)

```
1      # CLAUDE.md ? project memory / agent handoff
2
3      Local, AI-powered **badminton (? racket-sports) highlight indexer + rally detector**:
4      FastAPI + SQLite + ffmpeg + GPU CV/AI, with paid **Gemini** as the cloud AI. Sibling
5      repos: `sports-data-collector`, a VLC-based `rally-annotator`.
6
7      ## Read these first
8      - **`docs/README.md`** ? doc index. **`docs/CODE_MAP.md`** ? code navigation + data
9        contracts (the cold-start map; read before touching code).
10     - **Authoritative owned-model roadmap:** `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` (+
the
11     live `docs/NEXT_STEPS.md`). These SUPERSEDE the 2026-06-07 strategy docs below where
they
12     disagree (licensing, Gemma-4 stance, base model, conversational-QA).
13     - **Strategy (post-scaffolding):** `docs/COMMERCIALIZATION.md`,
`docs/PMF_MARKET_RESEARCH.md`,
14     `docs/PLATFORM_ARCHITECTURE.md`, `docs/DEFERRED_HARDENING_PLAN.md`. Also
`docs/ROADMAP.md`
15     (offline-segmenter narrative) and `docs/BACKLOG.md`.
16     - **Rally-detection quality track (design merged 2026-06-13, owner-gated, implementation
in progress):**
17     `docs/HOW_RALLY_DETECTION_WORKS.md` (2-layer mental model) ?
`docs/MULTI_SIGNAL_FUSION_PLAN.md`
18     (fuse shuttle + person + audio cues; the held-shuttle bug is the already-built
`rally_gate.py`
19     "Gate A" **not wired into production** ? cheapest win) ? `docs/EXPERIMENT_HARNESS.md`
(A/B + shadow
20     eval so cues land flag-gated/default-OFF with zero regression) ?
`docs/ROORA_LABELING_GUIDELINES.md`
21     (v8 vendor labeling spec). **`docs/NEXT_STEPS.md` has the prioritized pick-up for this
track.**
22     - **History:** `docs/archives/` ? ADRs (`decisions/DECISIONS.md`), research, and
23     **`checkpoints/`** (latest: `2026-06-strategy-foundation/`). **Archive policy:** only
move
24     **terminal-state** (DONE/REJECTED) work; live docs stay at `docs/` top level ? and
**before
25     archiving, HONESTLY update the doc first** (verify claims vs code ? flip status + a
completion
```

```

26     note ? update inbound refs ? keep the lineage ? then `git mv`), per the checklist +
living
27     lifecycle register in **`docs/DOC_STATUS.md`**.
28
29     ## Current state (June 2026)
30     - **Scaffolding complete**; strategy foundation + owned-model roadmap merged. **No
31     platform/owned-model implementation has started.** A 2026-06-10 audit-driven hardening
32     sweep landed (licensing gate, detector keying, re-ingest safety, config, eval gate,
API
33     validation, reliability, CI + test coverage) ? see git history.
34     - **Two tracks, both owner-gated:** the **committed next PLATFORM slice = async job
model**
35     (`DEFERRED_HARDENING_PLAN.md` #16, single-tenant); the **owned-model track** starts at
a
36     greenlit Phase-0 milestone (`OWNED_MODEL_IMPLEMENTATION_PLAN.md`). **Do not start
37     implementation without explicit owner approval.**
38
39     ## Architecture in one breath
40     - **Pluggable registries** (ABC + `Registry` singleton): segmenters / providers /
validators
41     / sports / annotations ? see `backend/pipeline/segmenters/base.py`. *Future:*
42     `StorageBackend` + `ComputeBackend` registries (PLATFORM §3).
43     - `video_id` = MD5 of the file ? `backend/utils/hashing.py::compute_video_id` (one
helper).
44     - Typed config = `backend/config/` (Pydantic). DB schema evolves via the `PRAGMA
45     user_version` migration ladder in `backend/infrastructure/database.py`.
46
47     ## Committed guardrails (non-negotiable)
48     - **Paid LLMs (Gemini/OpenAI/Anthropic) = inference / serving / fallback ONLY ? never a
49     training data source** (terms/copyright). Commercial-clean licensing for EVERY
dependency ?
50     code, data, AND weights: **MIT / Apache-2.0 / BSD only** (ratified 2026-06-09, owner-
agreed).
51     - **Owned-model base:** **Qwen3-VL (Apache-2.0)** primary, InternVL3 (MIT) fallback;
**Gemma-4
52     is AMBER / bench-only** (Apache grant likely but Prohibited-Use posture unconfirmed ?
counsel
53     needed; do NOT build on it without sign-off). **Reject Gemma 3** (viral license).
**Exclude
54     minors** from the training pool; per-user training data needs a separate explicit opt-
in.
55     - **v1 = rally + highlight + history + correction loop**; **gait/biometrics strictly
future**
56     (GDPR Art. 9).
57     - **Generic data schema**; `coach ? athlete` is an **optional overlay**, not baked in.
58     - **Compute = GCP ONLY (ADR-013, 2026-06-20):** GPU training + serving run on **GCP**
(`deploy/gcp/`;
59     Cloud Run+GPU for serving per ADR-012). **DigitalOcean is DROPPED for compute** (DO
GPU not enabled
60     on the account + not cleanly credit-funded; Paperspace is subscription-gated). Don't
re-evaluate
61     DO/Paperspace for compute without a new ADR; the `$200` DO credits go to **non-compute
only**. The
62     `deploy/digitalocean/` scripts are kept solely as provider-agnostic Linux-GPU setup
reused by GCP.
63
64     ## Workflow conventions
65     - **Branch off `master`; ONE PR per work item; NO PR stacking.** Don't open a PR unless
asked.
66     - Tests: `python run_all_tests.py` (offline, fully mocked). Python 3.10+ (the `trackers`
package requires ?3.10); `ffmpeg` required.
67     - Never push to the default branch directly; descriptive commit messages.
68     - **Substantial design/project work gets a tracked project doc** following
`docs/PROJECT_DOC_TEMPLATE.md` (status header · decisions-locked · progress tracker ·
definition-of-done; archived on completion). Precedents: `docs/RALLY_QUALITY_RESEARCH.md` §7,

```

```
`docs/GOLDEN_REGRESSION_FIXTURES.md`.
69 - **Launch / "bit-flip" PRs** (a default-flip that drives a tracked project to terminal
state) MUST, in the SAME PR, run the `docs/DOC_STATUS.md` $2 archival due-diligence on that
project's docs (verify-vs-code ? flip status + completion note ? update inbound refs ? keep ADR
lineage ? `git mv` to `docs/archives/` ? record in DOC_STATUS $3), and fix any internal-link
breakage the move causes. Enforced by `.github/PULL_REQUEST_TEMPLATE.md` + the `link-check` CI
job (`python -m tools.check_md_links`).
70 - ?? Hand the heaviest machine-side checks (real video, the full GPU/WSL/WASB suite) to
the owner,
71 who has the golden corpus + labelled videos; verify pure-logic paths locally.
72
```

4. user (2026-07-02T04:45:57.721Z)

480:### R2-5 [P1/S] ? badminton-highlight-indexer

5. user (2026-07-02T04:46:06.774Z)

```
480 ### R2-5 [P1/S] ? badminton-highlight-indexer
481 **Engine CLAUDE.md 'Current state' section is ~3 weeks false ? claims platform/owned-
model work 'has not started'**
482 - verdict: CONFIRMED
483 - reasoning: Personally reproduced against the working tree and git history. (1)
C:\Users\avidu\Projects\kshelsutra-guru\badminton-highlight-indexer\CLAUDE.md:30-31 states "***No
platform/owned-model implementation has started.**"; lines 34-37 say the "committed next
PLATFORM slice = async job model" and the owned-model track "starts at a greenlit Phase-0
milestone... Do not start implementation without explicit owner approval"; lines 41-42 call
`StorageBackend` + `ComputeBackend` registries "**Future:". git blame shows lines 30-37 were
last written 2026-06-10 (commit 62fb2fb) and never touched since, despite CLAUDE.md itself being
edited 6+ times after (e735fd4 06-13 ... af1ccbe 06-30 ? all other sections). (2) Current-tree
reality contradicts every one of those claims: backend/storage/ contains a StorageBackend ABC +
local/gcs/gdrive/gdrive_api/s3/capabilities implementations; backend/compute/base.py +
cloud_run.py exist (ComputeBackend); backend/api/job_worker.py + tests/test_async_jobs.py exist
? the "committed next slice" async job model already shipped (8bcf6ce #339 "make /api/compile
async (202 + poll)", 2026-06-22); backend/flywheel.py + backend/worker +
backend/eval/served_gate_a.py exist. Owned-model track: a555887 (2026-06-20) "M2 FINALIZED ?
Gen-0 weights/0.1.0-l4 pushed"; 4134721 archives DECOUPLED_COMPUTE as "DELIVERED (M0-M2 + M3.5
on GCP)"; 334ad0f (#336) M11 data-flywheel; b72e6de (#319) training worker fix. 314 commits
landed since 2026-06-13, through 2026-07-01 (e410136), including a recorded launch NO-GO
decision 2026-06-30 (4a7bfbe #404) ? i.e., the project reached launch-consideration, the
opposite of "not started". (3) No fix landed: git log -- CLAUDE.md shows the last commit
touching it (af1ccbe, 2026-06-30) added only the bit-flip-PR convention + Python 3.10 line; the
Current state section is untouched since 06-10. Bonus staleness in the same file: lines 18-19
claim rally_gate.py Gate A is "not wired into production" while backend/eval/served_gate_a.py
and c8a493d "served Gate-A litmus" (#401) indicate a served Gate-A path now exists.
484 - notes: Sharper failure scenario: CLAUDE.md is auto-loaded project memory for every
fresh agent session in this heavily agent-driven repo. A fresh session reads "No platform/owned-
model implementation has started" + "Do not start implementation without explicit owner
approval" and will (a) tell the owner the async job model is an un-started, gated future slice
and propose designing/implementing what already exists (backend/api/job_worker.py), (b) treat
StorageBackend/ComputeBackend as "Future" and re-design duplicates of backend/storage/ and
backend/compute/, (c) misreport project status upward (e.g., in handoffs/audits) as pre-
implementation when Gen-0 weights shipped and a launch NO-GO was formally recorded 06-30. This
also directly violates the workspace's own "keep it honest ? reflects real, shipped state"
convention. Correctly-scoped fix (S, one doc-only PR): rewrite CLAUDE.md "Current state" (lines
29-37) to reflect reality as of 2026-07-01 ? platform slice (async jobs, storage/compute
backends, capabilities/assets endpoints) SHIPPED; owned-model M0-M2+M3.5 DELIVERED on GCP, Gen-0
weights pushed; launch NO-GO 2026-06-30 with pointer to #404/docs; fix line 41-42 ("Future" ?
"implemented, see backend/storage/, backend/compute/"); fix lines 18-19 Gate-A "not wired" claim
against backend/eval/served_gate_a.py; re-point "committed next slice" at whatever
docs/NEXT_STEPS.md currently prioritizes. Not OWNER_GATED ? this is a factual truth-sync of docs
to shipped code, the same class as the repo's prior "audit truth-sync" commits (62fb2fb,
9736a3b); no spend or strategy decision required. Round-1 claim of "~3 weeks false" is accurate:
section frozen 2026-06-10, contradicting implementation landed from ~2026-06-20 onward.
485
```

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486     ### R2-6 [P1/M] ? sports-data-collector
487     **Golden-corpus registry rot: all 6 owned 'curated' videos point at a deleted absolute
local_path and the 4.9GB source videos are single-copy with no documented backup**
488     - verdict: CONFIRMED
489     - reasoning: Registry rot half CONFIRMED by direct observation. sports-data-
collector/data/sports_metadata.db (git-tracked; rows landed in commit 9e6f204 "#19 ingest the
n=6 owned golden corpus", DB last touched 9f1e792 with no path fix since): videos table has 8
status='curated' rows; the 6 owned ones (adarsh_avi_singles, kushagra_singles,
largetest_doubles, gbaaddy, testlarge_short, mahadevpura_singles; license_type=Proprietary,
is_commercial=1) all carry local_path under 'C:\Users\avidu\Projects\Annotation
Setup\Collect\Golden Labelled\*.MP4' ? that directory does not exist. The folder moved to
'C:\Users\avidu\Projects\khelsutra-guru\Annotation Setup\' whose 'Golden Labelled\' now contains
ONLY the six .rallies.csv label files; zero copies of the MP4s exist anywhere in the workspace
(searched). The committed export artifact data/eval_export/manifest.json repeats the same 6 dead
absolute paths. Code consequence at sports-data-collector/src/cli/curate.py:679-684: export of a
curated video with missing local_path emits md5=None plus a warning, and
src/core/manifests.py:23-26 documents md5 as "the indexer's join key" ? so any fresh export of
the entire owned golden corpus loses the deterministic MD5 join to badminton-highlight-indexer.
Data-loss half CONFIRMED as single-copy but the "no documented backup" wording is wrong: (a)
verified live via read-only 'gcloud storage ls' (2026-07-02): gs://khelsutra-rally-
corpus/videos/ has 10 objects and NONE of the six originals, while labels/ DOES hold all six
(2026-06-08_adarsh_avi_singles.rallies.csv etc.) and trajectories are in the bucket ? annotation
work-product is safe, source footage is not; (b) Google Drive golden folder (G:\My
Drive\Sharing\Annotation Videos\Golden Label Videos Request - June 15) holds only the 7 proxy
clips, not the original six; (c) the sole remaining copy is documented at F:\[Khelsutra] GoPro
Hero Black 12\KheISutra\Training\Golden Labelled\ per badminton-highlight-
indexer/scratch/copy_videos_to_gcs.ps1:31-36 and docs/DATA_IN_GCS.md:153 (§6 Phase-2) ? and the
F: drive is NOT currently mounted (only C: and G: exist), so even that copy's integrity is
unverifiable today, and no MD5s of the six were ever persisted anywhere
(eval_export/manifest.json predates the md5 field from PR #51; no proxy-registry rows).
490     - notes: Corrections to the round-1 claim: (1) Size is ~37.9 GB, not 4.9 GB ? per-clip
sizes annotated in scratch/copy_videos_to_gcs.ps1:31-36 (10.9+7.7+11+6.7+1.4+0.19 GB); the
~4.9-5.2 GB figure matches the three unrelated Roora pilot videos under 'Annotation
Setup/Archive/.../Annotation Videos/'. (2) "No documented backup" is false ? badminton-
highlight-indexer/docs/DATA_IN_GCS.md §5/§6/§8 explicitly documents the F: archive location and
a Phase-2 migration plan (tracker at doc line 246: 14/15 videos still unmigrated); but "single-
copy" stands: exactly one copy of the six source videos exists, on an intermittently-mounted
external HDD, unverifiable right now, with no persisted hashes. (3) Downgrade from potential P0
to P1 because labels + trajectories for all six ARE in gs://khelsutra-rally-corpus (verified),
so GPU-free eval/litmus and synth-train survive a drive failure; what would be permanently lost
is the proprietary source footage (no re-extraction via --trajectory-source detector, no re-
annotation, no re-rendering). Sharper failure scenarios: (a) F: HDD failure = permanent loss of
~38 GB of irreplaceable owned golden footage; (b) today, 'curate export' on the owned corpus
emits md5=None for all six (curate.py:679-684), silently breaking the indexer MD5 join contract
added in PR #51. Correctly-scoped fix (M, cross-repo): mount F:, verify + hash the six MP4s
(persist MD5s into the DB/manifest immediately ? S and decision-free), generate 1080p proxies,
upload via the existing scratch/copy_videos_to_gcs.ps1 or the rclone cloud path (Phase-2,
already planned), then repoint/annotate local_path (or add a gs:// video_uri column) in
sports_metadata.db and refresh data/eval_export/manifest.json. Partially OWNER_GATED sub-
decision only: DATA_IN_GCS.md §7.4 leaves proxies-only vs full-4K migration scope OPEN and the
upload is real (though tiny, cents/month) GCS spend; the hash-and-record step and the DB path
fix need no owner decision.
491

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6. user (2026-07-02T04:46:21.518Z)

```

master
e410136 api: /api/videos surfaces fps + resolution (from the stored ffprobe details) (#413)
81a78bf api: /api/videos returns video_id + a friendly match_name (fix disappearing name) (#412)
8b16cf4 docs: reflect today's storage endpoints + the corpus/user-video boundary (#411)

```

7. user (2026-07-02T04:46:28.111Z)

```

1     # Next steps (live) ? refreshed 2026-06-21
2

```

3 > ****How to read:**** this top section is the ****current cross-track state + prioritized pending list**** (refreshed 2026-06-21, replacing the stale 2026-06-13 header). The detailed ****owned-model & rally-quality track**** plan (2026-06-13) is ****preserved below, still valid**** ? it is now ***one of several*** active tracks. Live blow-by-blow: memory ``current-state.md``. Doc lifecycle/hygiene: ``docs/DOC_STATUS.md``.

4

5 **## ? Where we are (2026-06-21)**

6 ****DECOUPLED_COMPUTE delivered on GCP + archived**** (#270); ****compute = GCP only**** (ADR-013, DO dropped). ****`COMPUTE\_DECOUPLED\_SERVING`**** subproject spun up (#272, ****ADR-014****) ? one pure core + local/cloud profiles; ****owner M0 design sign-off pending****. ****Per-video workspace P0**** shipped (#264); ****doc-status register + archival due-diligence**** in place (#273). The ****khelsutra site + per-rally UI**** ship in parallel (sibling track). The ****rally-quality / owned-model**** track (below) is the core ML work, gated on ****growing the golden corpus****.

7

8 > ****? LAUNCH DECISION ? NO-GO (owner, 2026-06-30).**** The rally-detection product is ****not**** cleared for an external/commercial launch; we stay in ****development****. This is the expected default for the dev phase. Two unblockers gate a future GO: **** (1) **** set + meet the ****ship-gate target (#172)**** ? currently unset, needs corpus growth (and the R2 toLF1 ablation, item 8); **** (2) **** resolve the ****WASB shuttle-detector weight provenance**** (commercial-clean licensing). P2b (#396) was a do-no-harm quality increment, ****not**** a launch gate ? merging dev PRs ? launching. Re-decide only when both unblockers clear.

9

10 **## ? Prioritized pending ? all tracks**

11 ****P0 ? highest leverage / unblocks the most****

12 0. ****? [owner-priority, 2026-06-21] MAKE IT UI-DRIVEN + ALPHA-SHAREABLE ? "a few recordings lying around ? highlights, from a page, no SSH".**** Today the cloud flow is ****SSH/CLI-only**** (``docs/CLOUD_GPU_DRYRUN_GUIDE.md``); ****M12 does NOT fix it.**** Needs (a separate khelsutra ``web/`` SPA + deploy track, NOT an engine seam): **** (i) **** deploy the engine as a reachable ****service**** (M7 ? Cloud Run service, ? owner GCP spend for a ***persistent*** endpoint), **** (ii) **** the ****SPA ingest/progress screen**** (khelsutra ``web/`` B-P2: enter/upload video ? ``/api/process`` ? poll ``/api/jobs`` ? reel; ZERO engine code), **** (iii) **** flip ****M9 edge-auth ON**** (merged #290, default-OFF) to gate alpha tenants, **** (iv) **** a ****compute PICKER in the UI**** ("your machine vs our cloud", D13/D15). The engine seams make it ***possible***; the SPA makes it ***usable***. Full record: `[[ui-driven-cloud-serving-gap]]`.

13 0b. ****? [packaging track, IN PROGRESS ? 11 PRs merged 2026-06-21/22] Downloadable batteries-included app ? `docs/PACKAGING\_PLAN.md`.**** The LIGHT-tier engine is bundle-ready: ``pip install`` ? ``indexer --app`` serves the ****bundled SPA**** single-origin, ****torch-free****, with app-home state / ffmpeg resolver / DNS-rebinding guard / single-instance lock / safe DB upgrades (P0a?e, P1a/b, P2-launcher/P2a/P2d/P2e). ****? Next (fresh focused pushes):**** **** (1) **** P3 ****GPU validation + screencast**** (L4, ~\$0.50, DELETE-after, \$100 cap ? ``[[gcp-vm-always-delete]]``; ? native-WASB 0-rally tuned-config blocker), **** (2) **** P2b-install (uv/script + truly-local default config), **** (3) **** P2c first-run wizard (cross-repo khelsutra SPA from ``/api/capabilities``), then P3-cloud (Cloud Run) + P4 (ONNX/train/annotate = north star). Memory ``[[packaging-app-plan]]`` / ``[[next-session-packaging-app]]``.

14 1. ? ****`COMPUTE\_DECOUPLED\_SERVING` M0?M9 LARGELY SHIPPED**** (2026-06-21, ADR-014 ratified). M1?M4 (#279/#280/#282/#283), ****M5 #286, M6 #287, M8 #288, M9-auth #290**** all merged (default-OFF); the ****real M7 L4 run**** validated the worker (22 rallies, VM deleted, \$0). All \$8 gating Qs locked (D7?D17). ****Owner residual**** (not blocking Claude's default-OFF code): counsel ToS/DPA (M9-consent live + M11 live), real GCP prices+FX (M8 calibrate), Google OAuth app (M12 real), CF team-domain/AUD + ingress-lock; real cloud verification authorized within a ****\$100/?10k cap**** (D17).

15 2. ****[owner, ongoing] Grow the golden corpus**** ? multi-court badminton + ?3 matches/sport (TT first). The single highest-leverage move; the quality track ***and*** full-corpus training feed on it.

16 3. ****[owner + Claude . GPU] Full-corpus training (n?5)**** ? a real ``weights/<ver>`` (the ``0.1.0-l4`` bundle is n=2 plumbing-only).

17

18 ****P1 ? near-term, mostly Claude-doable (default-OFF, green-gated)****

19 4. ****[Claude] `COMPUTE\_DECOUPLED\_SERVING` batch ? REMAINING: M11 + M12**** (M5/M6/M8/M9-auth ? merged this session). ****M11**** = ``backend/flywheel.py`` capture?store?shadow-eval?goldenize plumbing reusing ``workspace.py::for_video`` + ``backend/eval/experiment.compare`` + ``backend/eval/promotion.py::promote_if_served_safe``, behind ``flywheel.*`=OFF` + a faked ``consent_attested`` (? privacy-sensitive ? give it fresh focus). **\*\*M12\*\*** = ``backend/storage/gdrive_api.py`` ``GDriveApiStorage(StorageBackend)`` + a ``gdrive-api://`` scheme (regex already allows the hyphen; add to ``_KNOWN`` + ``_backend_class`` + the StorageConfig

Literal) against an INJECTED fake Drive (mirror `gcs.py` DI); keep the mount-only `gdrive://` untouched. ? ****M9-consent cols = a new v7 migration**** (M8 took v6). M9-consent ToS/DPA text = owner/counsel.

20 5. ****[Claude] Per-video workspace P1?P6**** (P1 v6 DB migration ? P2 worker ? P3 per-video routing of `detector/+`tmp/` \*[the `output/chunks_` de-hardcode base = ? done via #282]* ? P4 ? P5 Launch-Report flip ? P6 GPU parity). ? P1's v6 migration coordinates with the M8 v6 cost-ledger migration ? author the version bump once.

21 6. ? ****Ops-Agent codified (#291)**** ? `deploy/gcp/create\_gpu\_vm.sh` (the one-command VM helper) always bakes the Ops-Agent startup-script; provision.sh grants the APIs+SA roles. PROVEN active on a fresh VM 2026-06-21. ([[gcp-ops-agent-enable]])

22 7. ****[owner] Set the ship-gate target (#172)**** ? needs corpus growth.

23 8. ****[owner-side GPU] R-series: promote-or-reject `R2` tolF1 as the gate**** (the ablation) ? needs GPU-extracted golden features.

24

25 ****P2 ? later / gated / deferred****

26 9. Rally-quality ****P3/P4**** (person-fusion LOVO, audio cue) ? needs GPU golden features.

27 9b. ? ****DONE (#396):**** the R4 density fix (finding B / P2b in `RALLY\_DETECTION\_FIXES\_PLAN.md`) is ****promoted to default-ON****. Golden-set A/B at the ****SERVED**** windowing showed tolF1 0.353?0.364 (?tolF1 +0.010, no regression), so `inactivity\_temporal\_density` default was flipped to True. (Landed flag-gated default-OFF in #394.) Fixed the sparse-tracking quirk where one fast sample after a detection gap held the rally tail open.

28 10. ****M0.4 scorer swap**** ? TemporalMaxer (per the 2026-06-17 external review).

29 11. ****Multisport**** (table-tennis first, then pickleball).

30 12. Serving ****M8/M9**** ? per-video cost ledger, edge auth, DPA/consent (within `COMPUTE\_DECOUPLED\_SERVING`; owner/legal-gated).

31 13. ****WASB-C3**** weight provenance (gates *sharing* a trained model) ? legal.

32 14. ****Rename repo off "badminton"**** (owner, BACKLOG).

33 15. Owner housekeeping: retire the shared ****DO droplet**** + ****DO token****; collector ****`md5`**** export (collector repo).

34 16. ****PMF field validation**** (owner-side, business track).

35

36 ****Doc-hygiene (tracked in `docs/DOC\_STATUS.md` \$4):**** ? #1 per-video reconcile + ? #2 this refresh; ****pending:**** field/PMF/Roora ? archives + a data-layer `docs/` dir (owner boundary confirm); confirm the COMPLETED top-level stubs; verify `HOSTING\_PLAN`/`UI\_PROPOSAL` currency.

37

38 ---

39

40 ****Owned-model & rally-quality track ? detail (2026-06-13 plan; still valid)**

41

42 ****Updated:**** 2026-06-13 (****rally-detection quality track****: fusion + experiment-harness design docs merged,

43 owner-gated ? see the dedicated section below, the desktop-agent pick-up). Prior: 2026-06-10 (audit?M0?Gemini-battery session; later same day the ****UI-alpha track opened****:

44 ADR-009 ratified ? KhelSutra Guru / khelsutra.guru, private `khelsutra` repo, local-first delivery ? see

45 `docs/UI\_ALPHA\_EXECUTION\_PLAN.md`; ****PR-1** async jobs (#16) MERGED 2026-06-11, PR #87**;

****PR-2**

46 upload/download MERGED 2026-06-11, PR #99** ? live-E2E-verified (3-part chunked upload MD5 byte-identity +

47 compile?browser-download); next: PR-3 identity. Same day: the ****8?9 quality sweep #90?#98 landed**** ? ruff/mypy/

48 coverage-floor CI lanes, SQLite conn-close fix, config unknown-key warning, hybrid-twin collapse into

49 `trajectory\_hybrid.py`, WSL tilde+abspath live-path fixes; suite 388 green; LOVO 0.626 re-verified exact).

50 ****Authoritative roadmap:****

51 `docs/OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md` (+ ADR-008 for topology). ****Latest blow-by-blow:**** memory

52 `current-state.md` / `session-handoff.md`. This file = the prioritized "what to do next," refreshed each session.

53

54 ****? Next-session / GPU-box pickup (2026-06-15)**

55 ****M0 in-container quick wins are COMPLETE:**** ship-gate provisional blocker (****#170****), train/eval split guard (****#171****), the ****doc-consistency sweep**** (****#174**** ? corrected 16

drifted docs vs the code), and the **P3** person-feature fusion litmus machinery is already present + suite-green** (`backend/eval/fusion\_compare.py` + `ablation.py`: person-driven ?F1 / bootstrap CI / paired sign test; the real-golden `skipif` litmus waits only on GPU-extracted data). ? CI is red repo-wide due to a **GitHub Actions minutes/billing outage**** (account-level, not code) ? work is local-gated + `--admin`-merged until Actions is restored (Settings ? Billing ? Actions minutes / spending limit).

56

57 **Immediate next step ? the real P3 LOVO measurement**** (this box has a GPU + `torch+cu124` + `cv2`; videos and trajectory CSVs are present, but the GPU-extracted golden features + `.rallies.csv` labels are not yet here):

58 1. **[owner / GPU box]**** `python -m backend.eval.fusion\_golden --manifest output/human\_lovo\_manifest.json --out-dir "%USERPROFILE%\OneDrive\rally-annotator\golden-data\features\2026-06-15-n6" ? produces `_golden_features.json` for the 6 golden videos. (GPU + WSL + videos + trajectory CSVs + `.rallies.csv` labels are owner-side only.)

59 2. Artifacts land in the **shared cloud folder**** `C:\Users\avidu\OneDrive\rally-annotator\golden-data\` (OneDrive auto-syncs both ways). Convention + workflow: **[GOLDEN_DATA_SHARING.md]**(data_pipeline/GOLDEN_DATA_SHARING.md)**; tracked in **#175**.

60 3. **[CPU dev/agent session]**** `python -m backend.eval.fusion\_compare --features-dir <shared>\features\2026-06-15-n6 --record` + `python -m backend.eval.ablation --features-dir <shared>\... --granularity group` ? real person-feature ?F1; copy the JSONs into `output/` to also pass the `skipif` litmus tests (`tests/test\_ablation.py::test\_litmus\_reproduces\_gate\_a`).

61

62 **Also queued (owner-gated / pending input):**

63 - Ship-gate target calibration ? **#172**** (owner decision; needs corpus growth).

64 - Tests reorganization ? blocked on the owner's grouping metric (provide it ? an agent will do it).

65 - Restore CI ? the Actions outage is account-level (billing/minutes), not a repo/config bug.

66

67 **Leaving (bulky / M2 ? no start without owner go):** ActionFormer (M0.4), event-index DB migration (M0.1), Gemini-silver purge (M0.3), cost-to-Gen-2 benchmark + ByteTrack deprecation, multisport, repo rename, PMF execution.

68

69 **Where we are (one paragraph)**

70 ADR-008 ratified ("repo split driven by productionization, not isolation"): training lives in-repo (`training/`)

71 behind a CI-enforced import wall; the featurizer is single-sourced (`backend/features/rally\_seq.py`, train==serve);

72 the **Gen-0 harness ships**** (`python -m training.gen0.harness --manifest output/human\_lovo\_manifest.json --floor`

73 `eval\_baselines/heuristic\_lovo\_n6.json --version <semver>`) and produced artifact v0.1.0 (LOVO **0.626****, floor passed,

74 sign test 5/6 wins p=0.109 ? honestly not significant, ship=False). The **n=6** owned corpus is in the collector**

75 (262 rallies, Proprietary, commercial export = owned data only after the ShuttleSet reclassification). The **Gemini**

76 battery is done** (see table) ? the moat is quantified.

77

78 **The quality table that now drives strategy (IoU 0.5 vs human golden labels)**

79 | Path | Mahadevpura (clean-ish) | Kushagra (multi-court) |

80 |---|---|---|

81 | Heuristic (velocity windowing) | 0.657 | 0.157 |

82 | **Gen-0 owned head**** (LOVO, n=6) | 0.744 | 0.561 |

83 | Two-stage WASB+Gemini refine | 0.83 | ? |

84 | **Gemini wrapper**** (`gemini-flash-latest`) | **0.93**** | **0.66**** |

85

86 **Reading:** clean footage is commoditized (serve it with Gemini for cents). **Multi-court** drops the commodity

87 wrapper to 0.66 ? that is the owned model's ship target** (its FPs are background-game pickups + dead-time merges,

88 the exact contrast-metric confound). Gen-0 is 0.10 behind with only n=6 ? a corpus-growth-sized gap, not an

89 architecture-sized one. Two-stage refine = the cost-efficient serving path for LONG tapes (uploads candidate clips

90 only); wrapper 503-flakiness (observed all session) makes the provider-fallback registry

load-bearing.

91 Baselines tracked in `eval\_baselines/` (heuristic_lovo_n6.json, gemini_wrapper_2026-06-10.json).

92

93 **## Rally-detection quality track (design merged 2026-06-13, owner-gated; implementation in progress)**

94 ****This is the track a desktop/local agent is slated to pick up.**** Design docs landed (PR #112 + session). All implementation is ****owner-gated**** (per explicit in the plan); each step = ONE PR off `master`.

95

96 ****Hardening loop (T1-T5 positive tests + audit + finale) COMPLETE (2026-06-14, PRs #156/#157/#160/#161/#163 + #165 + final records).**** See archived `docs/archives/CODE\_AUDIT\_AND\_TEST\_HARDENING-COMPLETED-2026-06-14.md` (Living Log with every pre/post/review) and `docs/archives/HARDENING\_LOOP\_HANDOFF-COMPLETED-2026-06-14.md` (full rules + STATE). Top-level now pointer stubs per #165 review. All 5 items + verification + archive done; clean slate.

97

98 - ****Docs:**** `docs/MULTI\_SIGNAL\_FUSION\_PLAN.md` (shuttle + person + audio, one fusion layer) .

99 `docs/EXPERIMENT\_HARNESS.md` (A/B + shadow + promotion gate, zero-regression) .

100 `docs/ROORA\_LABELING\_GUIDELINES.md` (v8) . `docs/HOW\_RALLY\_DETECTION\_WORKS.md` (2-layer mental model).

101 - ****Key finding (no re-investigation needed):**** the owner's "held-shuttle counted as a rally" bug is ****NOT** a missing capability** ? `backend/pipeline/detectors/rally\_gate.py` already implements the net-crossing ****"Gate A" (apply_gate(..., min_crossings=2))** that kills service-feed/held-shuttle windows, but it is ****only called from eval drivers**** (and there was no `min\_crossings` knob in `IndexingConfig`). ****Wiring Gate A into the live segmenter (now via fusion_hybrid) was the single cheapest, highest-confidence quality win.**** (P2 FusionSegmenter specifics: registry `fusion\_hybrid`, ****default-OFF****, seeds from substrate, persists `net\_crossings` + optional person features, optional served Gate A/B via config knobs; adversarially reviewed; suite green. See `tests/test\_served\_gate\_a\_regression.py` for the honest finding that on production windowing Gate A is neutral/negative in some cases, hence kept opt-in with knob for safety, and the leaderboard revert note.)

102 - ****Current status (2026-06-14):**** P1 (person cue extraction to reusable `PersonDetector`) + P2 (FusionSegmenter + Gate A signal + served handoff gating) + experiment harness P1 DONE (adversarial review + NO-REGRESSION; suite green). ****P3 next (this PR + follow-ups):**** learned fusion (person features ? harness `compare` LOVO lift on the 6 golden videos; eager-person-compute optimisation); then P4 audio via hit_detector.

103 - ****Pick-up order (each = ONE PR off `master`, owner approval first, flag-gated default-OFF, promote only on measured LOVO):****

104 1. (P2 first-step, largely done via fusion_hybrid) Wire Gate A into production served path + config knob (min_crossings); apply in hybrids at runtime. (Now available opt-in; served neutral on production windowing per litmus ? kept opt-in with knob for safety.)

105 2. P3 first step (this work item): harness `compare` LOVO litmus for person-on variant (players_in_court + two_sided + colocation etc. from `fusion\_features`) on the 6 golden; add the eager compute optimisation note/landing. Pure + harness (testable here).

106 3. P3/P4 continuation: audio cue integration + full learned promotion if lift proven.

107 - ****Discipline (from EXPERIMENT_HARNESS.md):**** every new cue ****flag-gated, default OFF****; promote to default ****only on measured LOVO lift**** through `run\_regression.py` (exit 0/1/3); pinned `CONTROL` ? rollback = config flip. Harness re-uses ~80% existing code.

108 - ? Gate A, person cue logic, harness, and pure fusion paths are testable here; the heavier real-video / golden LOVO / ablation confirmation waits on the GPU-extracted golden features + `rallies.csv` labels, which are not yet present in this checkout.

109 - ****External due-diligence review inputs (2026-06-17) ? PICK UP AFTER current ideas are exhausted; full details + citations in RALLY_QUALITY_RESEARCH.md §5.6.**** An owner-commissioned 31-reference review of the quality report ****validated**** the methodology and surfaced these ****future**** levers (research inputs, not active work, not new claims; verify citations before any submission):

110 1. ****"First-mile" paper framing (highest-value):**** stroke-classifiers (BST/DSTA) + the big datasets (ShuttleSet ~36k strokes, Fine-Badminton ~27k actions) all ****assume a pre-trimmed rally clip already exists**** ? we solve the ignored prerequisite. Pitch the paper as ****"the missing untrimmed-video preprocessing pipeline that lets BST/MonoTrack run on raw footage."****

111 2. ****Ground R2 in Action-Spotting / Precise-Event-Spotting**** (SoccerNet/TennisSet/T-DEED) ? turns R2 into a defensible publishable contribution.

112 3. ****Add domain baselines for any submission:**** MonoTrack + BST on ShuttleSet / Fine-

Badminton (we have only heuristic + Gemini today).

113 4. ****Prefer TemporalMaxer over ActionFormer**** for the M0.4 scorer swap (parameter-free max-pool, ~2.8× fewer GMACs, small-N robust) ? see "M0.4 next increment" below.

114 5. ****Court polygons ? pose ? the rally-END lever:**** masking on-court players can resolve *occluded end-of-rally states* (the #1 remaining gap vs Gemini), not just spectator noise.

115 6. ****Name the eval?serve lesson "pipeline distribution skew"**** ? a citable paper "lesson."

116

117 ****Discipline reminder:**** owner approval before starting any owner-gated item; one PR per step; babysit the PR (poll/fix/reply) until merge observed, *then* advance.

118

119 **## Prioritized next steps**

120 1. ****[owner, in progress] GROW THE GOLDEN CORPUS**** ? the single highest-leverage move; every lever below feeds on it.

121 Priority order for new videos: ****(a) multi-court badminton**** (the target distribution AND where the ship target

122 lives), ****(b) ?3 matches per new sport ? table tennis first**** (WASB transfers at F1 0.909; pickleball labels are

123 corpus-gold but not trainable until a clean MIT/Apache ball detector exists), (c) capture variety per

124 ``VIDEO_RECORDING_GUIDELINES.md``; ****adult sessions only for the training pool**** (consent guardrail). Per video:

125 label ? ``curate import-local --normalize --license Proprietary --fps <true fps>`` ? re-run the Gen-0 harness.

126 The paired sign test needs n: at n=10 with 9 wins, p?0.011 ? significance is reachable this month.

127 2. ****[owner decision] Set the ship-gate target**** ? now informed: floor = heuristic 0.480 (necessary), ****multi-court**

128 reference = wrapper 0.66** (the number that makes the owned model worth serving), clean-footage ceiling = 0.93

129 (not the target ? Gemini serves that tier). Suggested shape: "LOVO F1@tIoU0.5 ? 0.70 on multi-court folds with

130 paired-sign p<0.05 vs heuristic" ? owner to ratify/adjust.

131 3. ****M0.4 next increment (\$0/local):**** swap the Gen-0 LogReg for a lightweight temporal head (MIT, minutes on the

132 2070S ? NOT the heavyweight Gen-2 stack) inside the existing harness; per-video threshold calibration (the

133 LOVO-vs-oracle gap is visible per fold in the harness output). ****Prefer TemporalMaxer**** (parameter-free

134 max-pool, ~2.8× fewer GMACs, small-N robust) over ActionFormer/ASFormer per the 2026-06-17 external review

135 (``RALLY_QUALITY_RESEARCH.md`` \$5.6) ? better fit for the cheap/local/fast mandate.

136 4. ****Multisport thread:**** ingest Roora pickleball/TT CSVs (``import-local --normalize``); merged-prototype LOVO across

137 badminton+TT once TT has ?3 matches; identify a clean pickleball ball detector. ? TT-on-WASB serving stays gated

138 by C3 (provenance multiplies per sport).

139 5. ****PMF validation (cheaper than any engineering month):**** C13 academy interviews (+ the two added questions:

140 "multi-court footage nobody watches?" / "pay per-match to index it?"); camera pilot at the warmest 2-3 academies

141 (consent at capture; demo served via the Gemini path or human-polish ? the reel pattern exists:

142 ``output/MahadevpuraSingles_highlights.mp4``). ****Field kit COMPLETE (2026-06-12, owner ratified the**

143 physical-visit approach; reframed product-feedback-first per owner spec):**

144 ``docs/PRODUCT_FIELD_ASSOCIATE_FLYER.md`` (the recruiting flyer ? owner fills ?/contact placeholders and recruits) +

145 ``docs/FIELD_VISIT_PLAYBOOK.md`` (the hired associate's visit playbook: coach + student script, demo-after-Q6

146 rule, ground rules, Phase-2 record-and-show-back stretch goal) +

``docs/FIELD_VISIT_ANSWER_SHEET.md`` (per-visit

147 capture form incl. demo-reaction + student blocks) + ``docs/PMF_FIELD_SIGNALS.md`` (G/A/R anchors + ****pre-registered**

```

148         decision rules** ? PROCEED / PIVOT A-C / FALSIFIED thresholds fixed before the first
visit).
149     6. **Housekeeping:** ~~rotate the Gemini API key~~ **DONE 2026-06-10** (owner-verified
working; confirm old-key
150         deletion in AI Studio); **rename repo off "badminton"** (owner action,
151         in BACKLOG); collector=SoR confirmation; off-peak niceties: none pending ? the
battery completed.
152
153     ## Pointers / artifacts
154     - Roadmap: `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` · Topology: ADR-008
(`docs/archives/decisions/DECISIONS.md`) ·
155     Study: `docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md` · Quality history:
156     `docs/archives/quality-iterations/2026-06-multivideo-lovo/`.
157     - Harness: `training/gen0/harness.py` (one command ? train ? LOVO ? gate ? artifact
bundle; weights gitignored,
158     manifest reviewable). Featurizer: `backend/features/rally_seq.py` (FEATURE_SCHEMA v1).
159     - Corpus: collector DB (system-of-record, 6 matches/262 rallies) · eval manifest
`output/human_lovo_manifest.json`
160     (gitignored) · durable trajectories + Roora files: `C:\Users\avidu\Projects\Annotation
Setup\Collect\Trajectories\` ·
161     labels: `...\Collect\Golden Labelled\`.
162     - Gemini ops learnings: serial uploads (`indexing.ai_max_workers=1`) + chunk re-encode
163     (`indexing.ai_chunk_scale_width=1280`) + 6-attempt exponential generate retry ? a
cold/prepaid project burst-throttles
164     with a MISLEADING "credits depleted" 429; ramp gradually.
165     - ? No owned-model implementation beyond greenlit M0.x until owner greenlights the next
milestone. Committed next
166     PLATFORM slice = async job model (single-tenant).
167

```

8. user (2026-07-02T04:46:41.687Z)

```

backend/compute/:
__init__.py
__pycache__
base.py
cloud_run.py

```

```

backend/storage/:
__init__.py
__pycache__
base.py
capabilities.py
gcs.py
gdrive.py
gdrive_api.py
local.py
ref.py
s3.py
backend/api/job_worker.py
backend/eval/served_gate_a.py
backend/flywheel.py

```

9. user (2026-07-02T04:46:43.104Z)

```

af1ccbe chore(#399): launch-PR process hardening ? link-check CI, PR template, doc cross-links
(#402)
3624a8d feat(D13/P12): migrate ByteTrack to standalone trackers package (#386)
aa5e9db docs: add DOC_STATUS register + archival due-diligence convention (#273)
b05401b docs: drop DigitalOcean ? GCP is the sole compute stack (ADR-013) (#266)
9736a3b docs: re-ground stale doc claims to the current code (multi-agent audit) (#242)
1de2817 docs: freeze Golden Regression Fixtures plan + adopt tracked-project-doc template (#186)
e735fd4 docs: experiment-harness (A/B + shadow, no-regression) design + handoff updates
62fb2fb docs: sync to ratified decisions + repo reality (audit truth-sync) (#71)
3fd7232 docs: handoff-consistency sweep + strategy checkpoint + CLAUDE.md (#59)

```

8bcf6ce feat(api): make /api/compile async (202 + poll) to kill the slow-stitch 524 (#329)
(#339)
334ad0f feat(serving): M11 data-flywheel skeleton ? capture?shadow-eval?promote (default-OFF + consent faked-deny) (#336)
4a7bfbe docs: record the launch NO-GO decision (owner, 2026-06-30) (#404)
c8a493d fix(#398): served Gate-A litmus passes a typed IndexingConfig (not a dict) (#401)
a555887 docs(deploy): M2 FINALIZED ? Gen-0 weights/0.1.0-l4 pushed; both VMs deleted (\$0) (#267)
4134721 docs: archive DECOUPLED_COMPUTE ? DELIVERED (M0-M2 + M3.5 on GCP) (#270)

10. user (2026-07-02T04:46:52.336Z)

```
18:SCHEMA_VERSION = 9
83:    ``user_version`` stays un-bumped ? and the next open would re-run the same
102:        # Versioned migrations live here. PRAGMA user_version is the schema
105:        version = cursor.execute("PRAGMA user_version").fetchone()[0]
107:        # P2d upgrade-safety (no-op for a fresh DB or one already at SCHEMA_VERSION):
110:        if version > SCHEMA_VERSION:
113:            f"than this one (knows up to v{SCHEMA_VERSION}). Upgrade the app, or
point it at "
119:            if 0 < version < SCHEMA_VERSION:
148:            # future: if version < 10: cursor.execute("ALTER TABLE ..."); PRAGMA
user_version = 10
258:        cursor.execute("PRAGMA user_version = 1")
285:        cursor.execute("PRAGMA user_version = 2")
301:        cursor.execute("PRAGMA user_version = 3")
327:        cursor.execute("PRAGMA user_version = 4")
365:        CREATE UNIQUE INDEX IF NOT EXISTS idx_user_models_user_version
375:        cursor.execute("PRAGMA user_version = 5")
437:        cursor.execute("PRAGMA user_version = 6")
446:        cursor.execute("PRAGMA user_version = 7")
455:        cursor.execute("PRAGMA user_version = 8")
465:        cursor.execute("PRAGMA user_version = 9")
```

11. user (2026-07-02T04:46:53.903Z)

3a34c42 ci: add per-job timeout-minutes to every ci.yml job (hardening) (#382)
24178a7 docs(job_worker): document single-executor 02 risk + existing mitigations (#374)
de09ec8 refactor(main): app-factory + typed-config migration (D2/D3/P2/P3/P5/P7) (#370)
9329473 docs(cloudrun): note GOOGLE_CLOUD_PROJECT is required (run_job fails fast if unset)
(#345)
239c8eb fix(scripts): resolve gcloud via PATH in verify_compute_path so --verify-gcp works on
Windows (#344)
688d28f fix(compute): guard GoogleCloudRunJobClient against an unset GOOGLE_CLOUD_PROJECT (#342)
d92cc10 feat(serving): make the cloud compute-picker path work + provable (provenance +
skip_ai_handoff forward) (#341)
8bcf6ce feat(api): make /api/compile async (202 + poll) to kill the slow-stitch 524 (#329)
(#339)
819e395 feat(serving): M12 ? gdrive-api:// OAuth/API StorageBackend (D14 Northstar) (#338)
334ad0f feat(serving): M11 data-flywheel skeleton ? capture?shadow-eval?promote (default-OFF +
consent faked-deny) (#336)
6d3d054 feat(serving): SPA compute-picker ? per-request local/cloud override on /api/process
(item 3a) (#335)
e221cb9 docs(data): GCS corpus source-of-truth map + Phase-1 metadata migration (#316)
0ed4cc6 fix(segmenter): fail the job on an unrunnable config instead of a silent 0-rally success
(Finding B) (#311)
f01dc85 feat(db): v7 ? record the UI locale per job (PMF analytics) + count_jobs_by_locale
(#306)
c039b90 feat(pkg): P0e ? single-instance lock so a 2nd process can't corrupt in-flight job state
(default-OFF) (#298)
43eccc1 feat(serving): M8 ? per-job cost ledger + versioned pricebook + /cost API (default-OFF)
(#288)
41d45fd fix(server): drain async jobs under `python -m backend.main` (double-module hang) (#281)
ce22791 feat(serving): M1 ? deployment.profile + compute_target config (pure, additive, default-
OFF) (#279)
49c082a docs(serving): lock the 5 M0 gating decisions (D7-D12) + add the data-flywheel harness

```
(M11) (#277)
bd42250 docs: spin up COMPUTE_DECOUPLED_SERVING subproject + ADR-014 (#272)
4134721 docs: archive DECOUPLED_COMPUTE ? DELIVERED (M0-M2 + M3.5 on GCP) (#270)
b05401b docs: drop DigitalOcean ? GCP is the sole compute stack (ADR-013) (#266)
dc9111b docs(adr): ADR-012 ? GPU-native compute; TPU deferred; Cloud Run+GPU is the serving
model (#263)
380a4d0 docs(decoupled): narrow DECOUPLED_COMPUTE scope (M0?M2 + M3.5); defer M3/M4; DO=train,
GCP=serve (ADR-011) (#260)
38c2d0f feat(storage): gdrive:// reads a locally MOUNTED Google Drive (no API) (#258)
```

12. user (2026-07-02T04:47:38.335Z)

```
23:_KNOWN = ("local", "s3", "gcs", "gdrive", "gdrive-api")
27:class StorageStat:
37:class StorageRef:
69:     if scheme in _KNOWN:
```

13. user (2026-07-02T04:47:39.288Z)

```
1: ""The ``ComputeBackend`` ABC ? one interface for *where* the rally core runs.
3: COMPUTE_DECOUPLED_SERVING M6. Mirrors the ``StorageBackend`` ABC+factory idiom
15: from abc import ABC, abstractmethod
21: class ComputeJobSpec:
34: class ComputeResult:
50: class ComputeBackend(ABC):
```

14. user (2026-07-02T04:47:50.418Z)

```
__init__.py
__pycache__
_keyhint.py
base.py
fusion_hybrid.py
gemini.py
motion.py
tracknet_hybrid.py
trajectory_hybrid.py
wasb_hybrid.py
yolo_hybrid.py
backend/config/models.py:121:     threshold is raised (`min_crossings`/`min_players` default
0 = OFF). The court mask
backend/config/models.py:137:     min_crossings: int = Field(default=0, ge=0)
Binary file backend/config/__pycache__/models.cpython-313.pyc matches
```

15. assistant (2026-07-02T04:48:00.427Z)

All verified. Now creating the isolated worktree from origin/master.

16. user (2026-07-02T04:48:05.557Z)

```
Preparing worktree (new branch 'docs/claude-md-current-state-truth-sync')
branch 'docs/claude-md-current-state-truth-sync' set up to track 'origin/master'.
HEAD is now at e410136 api: /api/videos surfaces fps + resolution (from the stored ffprobe
details) (#413)
docs/claude-md-current-state-truth-sync
e410136 api: /api/videos surfaces fps + resolution (from the stored ffprobe details) (#413)
```

17. user (2026-07-02T04:48:12.526Z)

```
1      # CLAUDE.md ? project memory / agent handoff
2
3      Local, AI-powered **badminton (? racket-sports) highlight indexer + rally detector**:
4      FastAPI + SQLite + ffmpeg + GPU CV/AI, with paid **Gemini** as the cloud AI. Sibling
5      repos: `sports-data-collector`, a VLC-based `rally-annotator`.
```

```

6
7     ## Read these first
8     - **`docs/README.md`** ? doc index. **`docs/CODE_MAP.md`** ? code navigation + data
9       contracts (the cold-start map; read before touching code).
10    - **Authoritative owned-model roadmap:** `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` (+
the
11      live `docs/NEXT_STEPS.md`). These SUPERSEDE the 2026-06-07 strategy docs below where
they
12      disagree (licensing, Gemma-4 stance, base model, conversational-QA).
13    - **Strategy (post-scaffolding):** `docs/COMMERCIALIZATION.md`,
`docs/PMF_MARKET_RESEARCH.md`,
14      `docs/PLATFORM_ARCHITECTURE.md`, `docs/DEFERRED_HARDENING_PLAN.md`. Also
`docs/ROADMAP.md`
15      (offline-segmenter narrative) and `docs/BACKLOG.md`.
16    - **Rally-detection quality track (design merged 2026-06-13, owner-gated, implementation
in progress):**
17      `docs/HOW_RALLY_DETECTION_WORKS.md` (2-layer mental model) ?
`docs/MULTI_SIGNAL_FUSION_PLAN.md`
18      (fuse shuttle + person + audio cues; the held-shuttle bug is the already-built
`rally_gate.py`
19      "Gate A" **not wired into production** ? cheapest win) ? `docs/EXPERIMENT_HARNESS.md`
(A/B + shadow
20      eval so cues land flag-gated/default-OFF with zero regression) ?
`docs/ROORA_LABELING_GUIDELINES.md`
21      (v8 vendor labeling spec). **`docs/NEXT_STEPS.md` has the prioritized pick-up for this
track.**
22    - **History:** `docs/archives/` ? ADRs (`decisions/DECISIONS.md`), research, and
23      **`checkpoints/`** (latest: `2026-06-strategy-foundation/`). **Archive policy:** only
move
24      **terminal-state** (DONE/REJECTED) work; live docs stay at `docs/` top level ? and
**before
25      archiving, HONESTLY update the doc first** (verify claims vs code ? flip status + a
completion
26      note ? update inbound refs ? keep the lineage ? then `git mv`), per the checklist +
living
27      lifecycle register in **`docs/DOC_STATUS.md`**.
28
29    ## Current state (June 2026)
30    - **Scaffolding complete** ; strategy foundation + owned-model roadmap merged. **No
31      platform/owned-model implementation has started.** A 2026-06-10 audit-driven hardening
32      sweep landed (licensing gate, detector keying, re-ingest safety, config, eval gate,
API
33      validation, reliability, CI + test coverage) ? see git history.
34    - **Two tracks, both owner-gated:** the **committed next PLATFORM slice = async job
model**
35      (`DEFERRED_HARDENING_PLAN.md` #16, single-tenant); the **owned-model track** starts at
a
36      greenlit Phase-0 milestone (`OWNED_MODEL_IMPLEMENTATION_PLAN.md`). **Do not start
37      implementation without explicit owner approval.**
38
39    ## Architecture in one breath
40    - **Pluggable registries** (ABC + `Registry` singleton): segmenters / providers /
validators
41      / sports / annotations ? see `backend/pipeline/segmenters/base.py`. *Future:*
42      `StorageBackend` + `ComputeBackend` registries (PLATFORM §3).
43    - `video_id` = MD5 of the file ? `backend/utils/hashing.py::compute_video_id` (one
helper).
44    - Typed config = `backend/config/` (Pydantic). DB schema evolves via the `PRAGMA
45      user_version` migration ladder in `backend/infrastructure/database.py`.
46
47    ## Committed guardrails (non-negotiable)
48    - **Paid LLMs (Gemini/OpenAI/Anthropic) = inference / serving / fallback ONLY ? never a
49      training data source** (terms/copyright). Commercial-clean licensing for EVERY
dependency ?
50      code, data, AND weights: **MIT / Apache-2.0 / BSD only** (ratified 2026-06-09, owner-

```

agreed).

18. assistant (2026-07-02T04:49:00.626Z)

Let me verify referenced docs exist before writing, then make the three edits.

19. user (2026-07-02T04:49:04.242Z)

```
docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md
docs/PACKAGING_PLAN.md
tests/test_async_jobs.py
tests/test_served_gate_a_regression.py
```

20. assistant (2026-07-02T04:49:09.471Z)

Now the three edits.

21. user (2026-07-02T04:49:17.840Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-claudemd-r25/CLAUDE.md has been updated successfully. (file state is current in your context ? no need to Read it back)

22. user (2026-07-02T04:49:25.956Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-claudemd-r25/CLAUDE.md has been updated successfully. (file state is current in your context ? no need to Read it back)

23. user (2026-07-02T04:49:28.780Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-claudemd-r25/CLAUDE.md has been updated successfully. (file state is current in your context ? no need to Read it back)

24. assistant (2026-07-02T04:49:35.530Z)

Docs-only change ? no Python touched, so the relevant local check is the repo's markdown link-check CI job.

25. user (2026-07-02T04:49:38.617Z)

```
[check-md-links] clean ? no broken internal links in live docs.
exit=0
```

26. user (2026-07-02T04:49:44.312Z)

```
CLAUDE.md | 33 ++++++-----
 1 file changed, 20 insertions(+), 13 deletions(-)
docs/claude-md-current-state-truth-sync
e410136 api: /api/videos surfaces fps + resolution (from the stored ffprobe details) (#413)
```

27. user (2026-07-02T04:49:58.814Z)

```
[docs/claude-md-current-state-truth-sync 3b36f9f] docs(CLAUDE.md): truth-sync the stale Current-
state section to shipped reality (audit R2-5)
 1 file changed, 20 insertions(+), 13 deletions(-)
3b36f9f docs(CLAUDE.md): truth-sync the stale Current-state section to shipped reality (audit
R2-5)
```

28. user (2026-07-02T04:50:06.872Z)

```
1      <!-- Describe the change and link the issue it closes. -->
```



```

2
3     ## Summary
4
5
6
7     ## Checks
8     - [ ] `python run_all_tests.py` green; `ruff check .` + `mypy` clean; coverage not
lowered.
9     - [ ] Internal markdown links valid (`python -m tools.check_md_links`).
10
11     ## Launch / "bit-flip" PRs only
12     A PR that flips a default / "launch bit" so a tracked project reaches terminal state
MUST also run
13     the `docs/DOC_STATUS.md` §2 archival due-diligence on that project's docs, **in this
same PR**:
14     - [ ] Verified each doc's claims against the code (tagged load-bearing facts
`[verified]`).
15     - [ ] Flipped the doc Status ? DELIVERED/DONE/REJECTED + a completion note (what
shipped, PR, paths).
16     - [ ] Updated all inbound references (`docs/README.md`, `CLAUDE.md`,
`docs/NEXT_STEPS.md`, cross-doc links).
17     - [ ] Kept the ADR forward/back lineage.
18     - [ ] `git mv` the terminal docs to `docs/archives/?` and recorded the move in
`DOC_STATUS.md` §3.
19     - [ ] Fixed any internal-link breakage from the move (`link-check` CI job green).
20
21     _(Not a launch/bit-flip PR? Delete this section.)_
22

```

29. user (2026-07-02T04:50:17.910Z)

```

remote:
remote: Create a pull request for 'docs/claude-md-current-state-truth-sync' on GitHub by
visiting:
remote:      https://github.com/Khelsutra/badminton-highlight-indexer/pull/new/docs/claude-md-
current-state-truth-sync
remote:
branch 'docs/claude-md-current-state-truth-sync' set up to track 'origin/docs/claude-md-current-
state-truth-sync'.
To https://github.com/Khelsutra/badminton-highlight-indexer.git
* [new branch]      docs/claude-md-current-state-truth-sync -> docs/claude-md-current-state-
truth-sync

```

30. user (2026-07-02T04:50:42.832Z)

<https://github.com/Khelsutra/badminton-highlight-indexer/pull/414>

31. user (2026-07-02T04:51:09.918Z)

Structured output provided successfully